```
wsumieniewiemdupa
kamilmarchewka@lima-MDO-server:/Users/kamilmarchewka$ sudo apt update && sudo apt install git -y
Hit:1 http://ports.ubuntu.com/ubuntu-ports questing InRelease
```

```
See 'git help git' for an overview of the system.
kamilmarchewka@lima-MDO-server:/Users/kamilmarchewka$ git --version
git version 2.51.0
```

Tworzenie kluczy ssh:

Bez hasła:

```
kamilmarchewka@lima-MDO-server:/Users/kamilmarchewka$ ssh-keygen -t ed25519 -f ~/.ssh/id_github_open
-N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/kamilmarchewka.linux/.ssh/id_github_open
Your public key has been saved in /home/kamilmarchewka.linux/.ssh/id_github_open.pub
The key fingerprint is:
SHA256:wB2+71gxJIgu2rU13dXIQYhNvJXVEdx8uMkF+/n2FEM kamilmarchewka@lima-MDO-server
The key's randomart image is:
+--[ED25519 256]--+
|        .=.ooo+B+|
|     o +..+.o+.o*|
|    . + + .o+.oE.|
|   .   o =..  =..|
|  . o o S +    +.|
| o o o . . o    +|
|. . .     o     +|
|         +     o.|
|        . .     .|
+----[SHA256]-----+
```

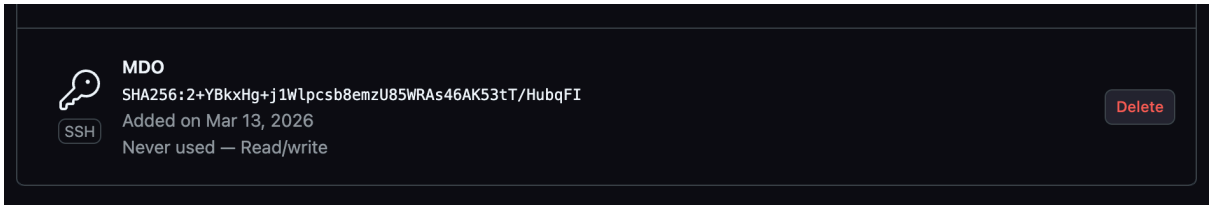
Z hasłem: (Qwerty123!@#)

```
+----[SHA256]-----+
kamilmarchewka@lima-MDO-server:/Users/kamilmarchewka$ ssh-keygen -t ed25519 -f ~/.ssh/id_github_secur
e
# Tutaj terminal poprosi Cię o wpisanie "passphrase" — podaj bezpieczne hasło.
Generating public/private ed25519 key pair.
Enter passphrase for "/home/kamilmarchewka.linux/.ssh/id_github_secure" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kamilmarchewka.linux/.ssh/id_github_secure
Your public key has been saved in /home/kamilmarchewka.linux/.ssh/id_github_secure.pub
The key fingerprint is:
SHA256:2+YBkxHg+j1Wlpcsb8emzU85WRAs46AK53tT/HubqFI kamilmarchewka@lima-MDO-server
The key's randomart image is:
+--[ED25519 256]--+
|      ...    ..  |
|     .   .. o .. |
|      . .. o o.  |
|    ... .o o.. . |
|    .+ .S.= +   .|
|     .o. BE+ .  +|
|      ..=o+.o ++.|
|      ..++ o.B.o.|
|       . oo.+o=o.|
+----[SHA256]-----+
```

```
MDO-server
kamilmarchewka@lima-MDO-server:/Users/kamilmarchewka$ ls -l ~/.ssh
total 20
-rw------- 1 kamilmarchewka kamilmarchewka  86 Mar 13 08:17 authorized_keys
-rw------- 1 kamilmarchewka kamilmarchewka 432 Mar 13 08:24 id_github_open
-rw-r--r-- 1 kamilmarchewka kamilmarchewka 112 Mar 13 08:24 id_github_open.pub
-rw------- 1 kamilmarchewka kamilmarchewka 484 Mar 13 08:25 id_github_secure
-rw-r--r-- 1 kamilmarchewka kamilmarchewka 112 Mar 13 08:25 id_github_secure.pub
```

Dodanie klucza publicznego do githuba:

```
+----[SHA256]-----+
kamilmarchewka@lima-MDO-server:/Users/kamilmarchewka$ cat ~/.ssh/id_github_secure.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAID5QvtTDBwD3Nsld3MMqu53GQ4pyMm1XksAwfczwBIUC kamilmarchewka@lima-
MDO-server
```



Klonowanie repozytorium:

Https:

```
kamilmarchewka@lima-MDO-server:~$ git clone https://github.com/InzynieriaOprogramowaniaAGH/MDO2026_ITE.git
Cloning into 'MDO2026_ITE'...
remote: Enumerating objects: 2085, done.
remote: Counting objects: 100% (423/423), done.
remote: Compressing objects: 100% (345/345), done.
remote: Total 2085 (delta 86), reused 266 (delta 55), pack-reused 1662 (from 2)
Receiving objects: 100% (2085/2085), 67.50 MiB | 36.55 MiB/s, done.
Resolving deltas: 100% (251/251), done.
kamilmarchewka@lima-MDO-server:~$ ls
MDO2026_ITE
```

SSH:

```
  GNU nano 8.4                    /home/kamilmarchewka.linux/.ssh/config *
Host github.com
    HostName github.com
    User git
    IdentityFile ~/.ssh/id_github_secure
    IdentitiesOnly yes
```

```
kamilmarchewka@lima-MDO-server:~$ chmod 600 ~/.ssh/config
```

```
kamilmarchewka@lima-MDO-server:~$ ssh -T git@github.com
Enter passphrase for key '/home/kamilmarchewka.linux/.ssh/id_github_secure':
Hi kamilmarchewka! You've successfully authenticated, but GitHub does not provide shell access.
```

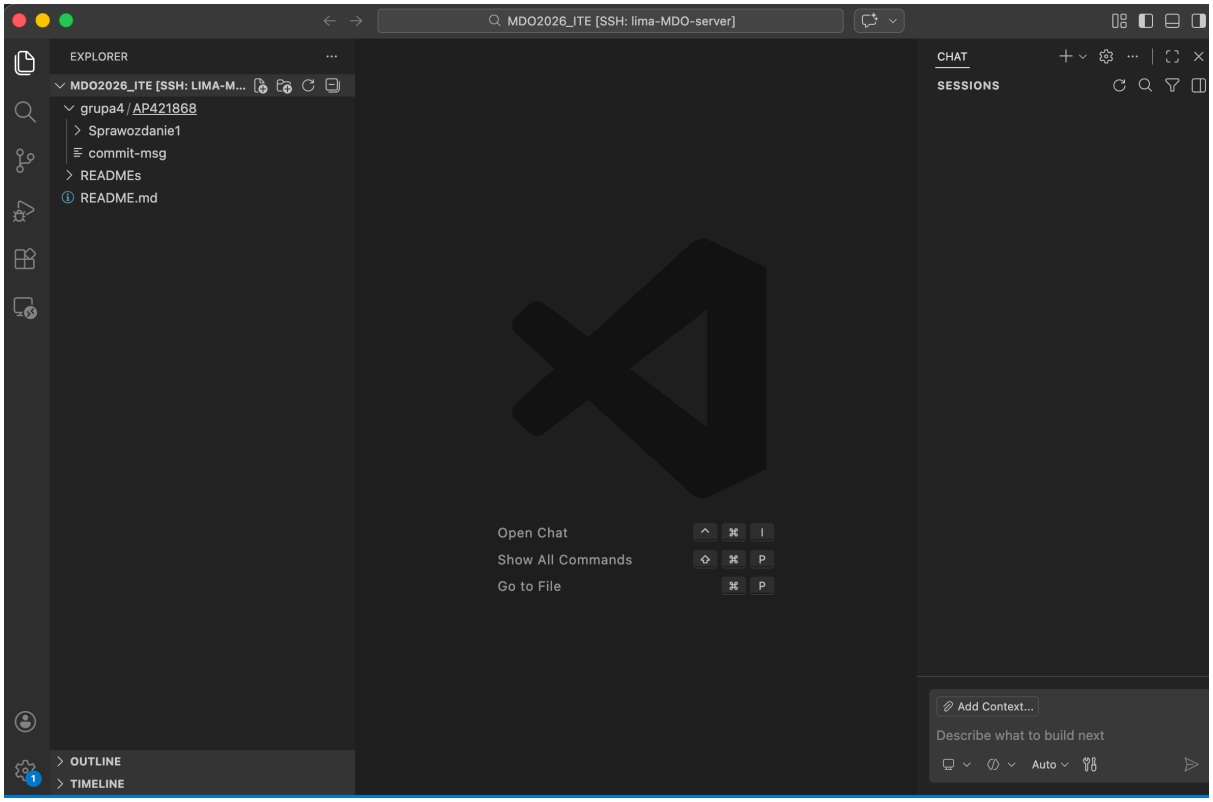
```
kamilmarchewka@lima-MDO-server:~$ git clone git@github.com:InzynieriaOprogramowaniaAGH/MDO2026_ITE.git
Cloning into 'MDO2026_ITE'...
Enter passphrase for key '/home/kamilmarchewka.linux/.ssh/id_github_secure':
remote: Enumerating objects: 2097, done.
remote: Counting objects: 100% (435/435), done.
remote: Compressing objects: 100% (351/351), done.
remote: Total 2097 (delta 89), reused 277 (delta 59), pack-reused 1662 (from 2)
Receiving objects: 100% (2097/2097), 67.50 MiB | 13.45 MiB/s, done.
Resolving deltas: 100% (254/254), done.
```

Łączenie z vs-code przez Remote – SSH

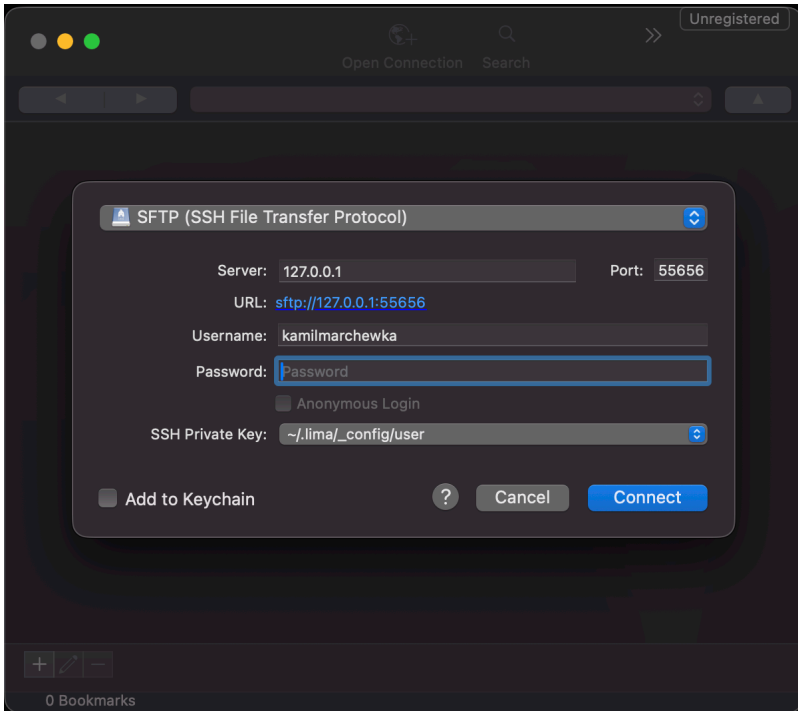
```
[~ » limactl show-ssh --format config MDO-server                    kamilmarchewka@macbookair ]
WARN[0000] `limactl show-ssh` is deprecated. Instead, use `ssh -F /Users/kamilmarchewka/.lima/MDO-ser
ver/ssh.config lima-MDO-server`.
Host lima-MDO-server
  IdentityFile "/Users/kamilmarchewka/.lima/_config/user"
  StrictHostKeyChecking no
  UserKnownHostsFile /dev/null
  NoHostAuthenticationForLocalhost yes
  PreferredAuthentications publickey
  Compression no
  BatchMode yes
  IdentitiesOnly yes
  GSSAPIAuthentication no
  Ciphers "^aes128-gcm@openssh.com,aes256-gcm@openssh.com"
  User kamilmarchewka
  ControlMaster auto
  ControlPath "/Users/kamilmarchewka/.lima/MDO-server/ssh.sock"
  ControlPersist yes
  Hostname 127.0.0.1
  Port 49332
```

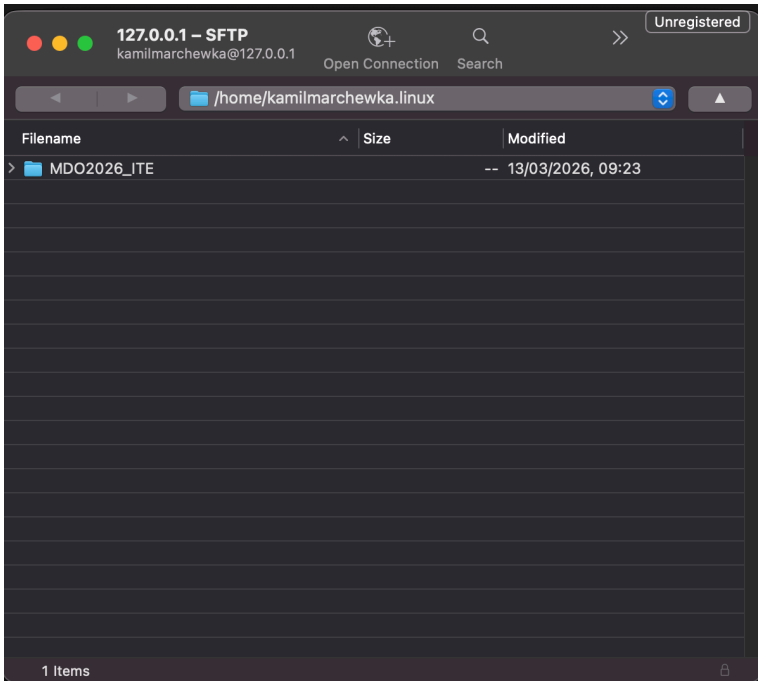
Dodanie konfiguracji do vs code

```
config
Users > kamilmarchewka > .ssh > config
2   Host alias
5
6   Host lima-MDO-server
7     IdentityFile "/Users/kamilmarchewka/.lima/_config/user"
8     StrictHostKeyChecking no
9     UserKnownHostsFile /dev/null
10    NoHostAuthenticationForLocalhost yes
11    PreferredAuthentications publickey
12    Compression no
13    BatchMode yes
14    IdentitiesOnly yes
15    GSSAPIAuthentication no
16    Ciphers "^aes128-gcm@openssh.com,aes256-gcm@openssh.com"
17    User kamilmarchewka
18    ControlMaster auto
19    ControlPath "/Users/kamilmarchewka/.lima/MDO-server/ssh.sock"
20    ControlPersist yes
21    Hostname 127.0.0.1
22    Port 49332
```

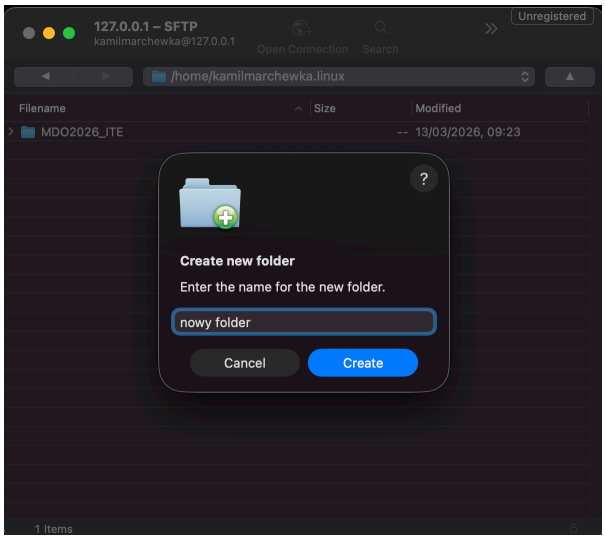


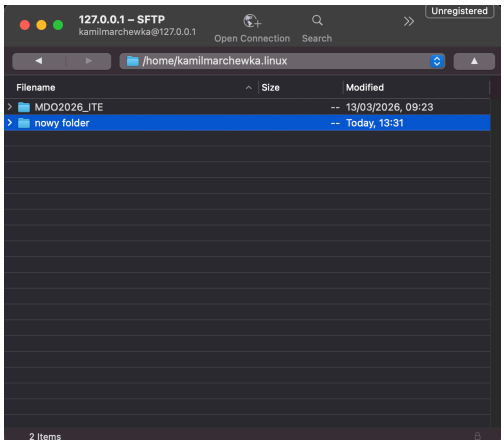
Konfiguracja klienta SFTP (Cyberduck)





Dodawanie plików:





Praca z gitem:

Przełączenie na gałąź grupy

```
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE$ git checkout grupa4
fatal: 'grupa4' could be both a local file and a tracking branch.
Please use -- (and optionally --no-guess) to disambiguate
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE$ ls
README.md  READMEs  grupa4
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE$ git checkout -b grupa4 --track origin/grupa4
branch 'grupa4' set up to track 'origin/grupa4'.
Switched to a new branch 'grupa4'
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE$
```

```
Switched to a new branch 'grupa4'
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE$ git checkout -B KM419688
Switched to a new branch 'KM419688'
```

```
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE$ cd grupa4
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE/grupa4$ ls
AN420700  BW414729  GN421256  MN420239  SM419482  TN420202
AP421868  DM421198  KM419774  MO420227  TM424276
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE/grupa4$ mkdir KM419688
```

Hook

```
git_commit_msg_hook.sh
#!/bin/bash

COMMIT_MSG_FILE=$1
COMMIT_MSG=$(head -n 1 "$COMMIT_MSG_FILE")
MY_ID="KM419688"

if [[ ! $COMMIT_MSG =~ ^$MY_ID ]]; then
    echo "======================================================"
    echo "BŁĄD: Commit zablokowany przez Git Hooka!"
    echo "Wiadomość musi zaczynać się od: $MY_ID"
    echo "======================================================"
    exit 1
fi

exit 0
```

Skopiowanie I nadanie uprawnień

```
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE/grupa4/KM419688$ ls ../../.git
FETCH_HEAD  ORIG_HEAD  description  index  logs     packed-refs
HEAD        config     hooks        info   objects  refs
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE/grupa4/KM419688$ cp git_commit_msg_hook.sh ../../.git/hooks/com
mit-msg
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE/grupa4/KM419688$ chmod +x ../../.git/hooks/commit-msg
```

Sprawdzenie działania

```
kamilmarchewka@lima-MDO-server:~/MDO2026_ITE/grupa4/KM419688$ git commit -m "text"
=====================================================
BŁĄD: Commit zablokowany przez Git Hooka!
Wiadomość musi zaczynać się od: KM419688
=====================================================
```